



Hypothesis Testing

Key terms

KSBs: K13, S11, S52, B1

hypothesis testing, null hypothesis, alternative hypothesis, statistical significance, critical value, critical region, one-tailed, two-tailed

6.1 Hypothesis Testing

Hypothesis testing is the process of applying statistical tests to experimental situations. An intervention, which it is suggested might cause a *change in the situation*, is compared to the state without intervention - and statistical analysis can tell us whether there is a *significant change*.

Definition 6.1 - Null Hypothesis and Alternative hypothesis

A **hypothesis** is a statement about the effect of an intervention on a situation, which you wish to test in order to *accept or reject it*.

The **null hypothesis**, denoted H_0 , is the statement encapsulating the situation in which there is *no effect from the intervention*. In the absence of sufficient evidence, we assume the null hypothesis is true *unless proven otherwise*.

The **alternative hypothesis**, denoted H_1 , is the statement we wish to test, and describes a *different outcome to the null hypothesis*.

Many different types of statistical test can be used, and we can compare many different things, including whether statistics about a set of data *have changed from previous measurements*, what the distribution of the data looks like, or whether a sample *was drawn from* a given distribution.

Example 6.1.

Suppose a company is considering putting a new type of coating on bearings that it produces. The mean wear life with the current coating is known to be *1000 hours*. The company would not want to make any (costly) changes unless evidence strongly suggested *it would improve the mean wear life*.

The hypothesis we would seek to test in this case would be that *the new mean wear life is greater than 1000 hours*.

If we denote the new mean wear life by μ_1 , we would write

$$H_0: \mu_1 = 1000 \quad H_1: \mu_1 > 1000$$

Exercise 6.1.

For each of the experiments described below, state the null hypothesis and the alternative hypothesis.

- A runner wishes to improve her 100m time, which is normally distributed with mean μ_0 . She trains intensively for a week, and measures her time for running 100m over 5 days.
- A group of 46 individuals with clinical depression were randomly assigned to either treatment as usual, or sessions playing a computer game. Participants performed a neuropsychological assessment which produced a score. The scores were then compared between the groups. Neuropsychological assessment scores are normally distributed, and higher scores on the test correspond to better mental health.
- A football fan wishes to determine if there is a biased pattern in the times at which yellow cards are given out during a match, or whether they follow a normal distribution. They watch all the games in a Premier League season and record the number of yellow cards given out during 10 minute intervals of each match.
- A computer gamer achieves a score of 40,000 points on a difficult game, which is a new world record. The scores for the game are normally distributed, and the adjudicator for the world record wishes to determine whether or not the player was cheating.

6.2 Experimental design

When we design experiments, we usually have a few common features.

- An independent variable, which you can control the value of; this could be a *numerical value, or a yes/no.*
- A dependent variable, which we can measure *having changed the independent variable*
- A sample of data to perform the test on. This sample *must be random.*

In order to make sure the effect you're measuring is due to the variable you're changing, it's often useful to apply the change to a *random subset of the sample*, and leave the rest unchanged - called a **control group**. This allows you to measure *the difference made by the intervention* more clearly.

It's also important, if possible, only to alter *one variable at a time* - otherwise, any measured change in the results could be due to any of the things you've changed, with no way to tell.

It may be the case that **you aren't in control of the exact parameters of the experiment**: you may be performing tests on *data someone else has previously gathered*, or you may not have the ability to collect a *true random sample* or establish a *control group*. If you can obtain information about the source of the data and how it was collected, you can take this into account when analysing the data.

6.3 Statistical Significance

If we measure a change following the intervention, it isn't necessarily the case that this change was *caused by the intervention* - natural variation in the variable could coincide with the intervention, in a way that might *look like a meaningful change.*

If data is drawn from an underlying *normally distributed dataset*, we would expect the values to vary

- with most values *close to the mean*, but some rarer examples will be *from further away*.

We use the concept of **statistical significance** to measure the difference between a *random fluctuation*, and something which could be considered an *actual response to the intervention*.

Definition 6.2 - p -value

The p -value is the probability of obtaining test results *at least as extreme* as the result actually observed, under the assumption that *the null hypothesis is correct*. A very small p -value means that such an extreme observed outcome would be *very unlikely* under the null hypothesis.

Example 6.2.

Muriel Bristol claims she can detect whether a cup of tea has been made by *pouring milk into the cup first or second* by tasting it. Her colleague prepares *8 cups of tea*, *4 of which* had milk poured first, and the other *4* had tea poured first.

There are *70* ways to choose 4 objects from a set of 8, so *70* possible ways the cups could have been arranged. Muriel *identified them all correctly* and so the probability of this happening by chance is *1 in 70* or $0.01428 = 1.42\%$.

In this experiment, the null hypothesis is that *Muriel is answering randomly* and has no way to tell if the milk was poured in first. Since this result was *extremely unlikely* to have occurred by chance, we can *reject the null hypothesis* and promote the alternative hypothesis - that Muriel has a special ability to detect whether tea or milk was poured first.

Definition 6.3 - Significance Level

The **significance level** is the value we have decided the p -value needs to be *smaller than* in order to reject the null hypothesis; conventionally, significance is set at *5%, or 0.05* .

In our earlier example, $1.4\% < 5\%$ so we can say that the result is *statistically significant*.

Note

5% is the standard significance level used across most of science and data analysis. There may be situations where other values are used, for specific types of scientific experiment - but generally 0.05 is standard.

Definition 6.4 - Test statistic

A **test statistic** is a single value obtained from a dataset in order to *perform a hypothesis test*.

Unlike a *descriptive statistic such as the Mean*, test statistics do not necessarily tell us much about the data, but they are instead used to test for significance.

In the examples we'll be looking at today which involve the normal distribution, the test statistic is the *Z-score* and this type of testing is called a *Z-test*. For other types of hypothesis test we will see later, the test statistic is calculated differently and has a different name.

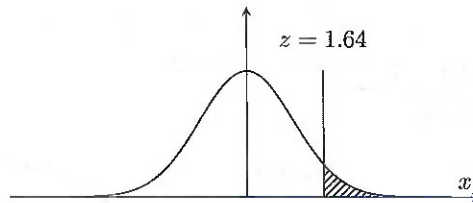
Definition 6.5 - Critical Value/Critical Region

For the significance level we have chosen, we can calculate the **critical value**, which is the value of the test statistic for which the p -value is **exactly at the significance level**.

The **critical region** is the range of values the test statistic can take for which the p -value is **lower than the significance level**, so we can **reject the null hypothesis**.

The critical value for a p -value of 5% will be the value of the test statistic for which the probability is **exactly 5% or 0.05** (or, the probability of it not happening is exactly **95% or 0.95**).

For the normal distribution, by looking it up in the table, we see a probability of 95% corresponds to a z -score of **1.64**, and anything to the right of this (with a z -score **greater than 1.64**) will be in the critical region.



If we are seeking the value of the test statistic to be **larger than the critical value**, we call this the **upper tail** of the distribution (the right-hand end of the diagram), and if we seek a value **smaller than the critical value**, we examine the **lower tail** (on the left of the diagram).

Example 6.3.

A factory makes widgets whose weights follow the distribution $N(155, 15^2)$. On visiting the factory, you find a stray widget on the floor, and want to know where it has come from. It may have either **fallen off the production line**, or have been **dropped by some visitors** who were in today from the Large Widget Factory down the road, whose widgets are **larger than the ones produced here**.

H_0 : the widget you have found was from the production line

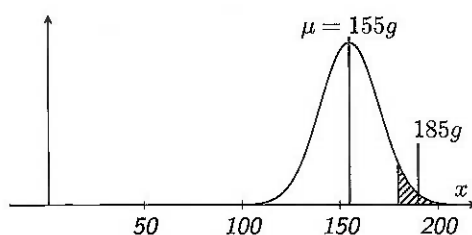
H_1 : the widget was dropped by one of the visitors.

The widget is weighed, and it is found to have a weight of 185g. The z -score (test statistic) for this value is

$$z = \frac{x - \mu}{\sigma} = \frac{185 - 155}{15} = 2$$

Looking this up in the table gives a p -value of $1 - 0.9772 = 0.0228 = 2.2\%$. This is **below the significance level of 5%**, therefore we might choose to **reject the null hypothesis**, and assume the widget was from the other factory.

We can also note that a z -score of **2** is greater than the critical value of 1.64, and **reject the null hypothesis on this basis**.



For this distribution, the weight corresponding to the critical value can be found by substituting back into the z-score calculation:

$$1.64 = \frac{x - \mu}{\sigma} = \frac{x - 155}{15}$$

$$1.64 \times 15 = x - 155 \Rightarrow (1.64 \times 15) + 155 = x \Rightarrow x = 179.6$$

So if the widget's weight is greater than 179.6g we know we can reject the null hypothesis.

Note

It is important to remember that the probability we are calculating is not the probability that (e.g. in this case) the widget is from the production line; it is the probability that if it were (if the null hypothesis is true), we would see a result this extreme.

6.3.1 Testing the distribution

As well as testing a specific value for whether it plausibly comes from a set with a given distribution, we can test whether the *distribution of a population has changed*, given an observation you know is from the set. This uses the *sampling distribution of the mean*.

Example 6.4.

A farmer has been monitoring the amount of lead in one of her fields next to a factory over many years. Over time, she has found the mean density of lead to be 67mg/kg with standard deviation 25mg/kg. Let X be a variable representing the amount of lead contained in the field. We have $X \sim N(67, 25^2)$.

After the factory closed, the farmer took 150 samples of soil, randomly selected from different parts of the field, and found the mean density of lead to be 62mg/kg. We assume that the amount of lead follows a normal distribution with standard deviation 25mg/kg.

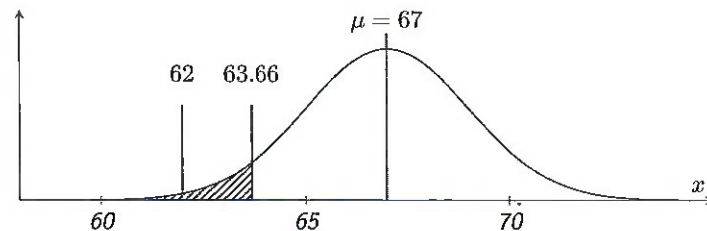
Since we are measuring a change in the mean, we can use the *distribution of sample means for $\bar{X}$* , which for samples of size 150 will be given by

$$\bar{X} \sim N\left(67, \frac{25^2}{150}\right)$$

We knew that the mean was 67mg/kg before the factory closed, so there's no reason to think it has changed. It has been suggested that the mean may have decreased, and that is what we would like to test.

H_0 : The mean has not changed, so $\mu = 67$

H_1 : The mean has decreased, so $\mu < 67$.



This is a lower tail test, so for a 5% significance level, the z-score needs to be in the critical region of -1.64 or less to reject the null hypothesis. For our sample, $\bar{x} = 62$. The z-score for this value can be calculated as:

$$z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{62 - 67}{\frac{25}{\sqrt{50}}} = -2.45$$

This is in the critical region, so we can reject the null hypothesis and accept the hypothesis that the amount of lead has decreased.

A z-score of -2.45 corresponds to a probability of $1 - 0.9929 = 0.0071$ or 0.7% which is lower than the significance level of 5%.

For a significance level of 5%, we can calculate the density corresponding to the critical value as:

$$1.64 = \frac{x - \mu}{\frac{s}{\sqrt{n}}} = \frac{x - 67}{\frac{25}{\sqrt{50}}}$$

$$1.64 \times 2.041 = x - 67 \Rightarrow (1.64 \times 2.041) + 67 = x \Rightarrow x = 70.34$$

This is the value to the right of the mean, and to find the equivalent value to the left we can calculate $(70.34 - 67)$ and subtract this from 67: $67 - (70.34 - 67) = 63.66$.

If the density of the sample is below this value, we can reject the null hypothesis.

Exercise 6.2.

- A biologist studying elk measures the weights of elk in the park where they work over the course of a year, to make sure they are getting enough food. The biologist finds that based on a sample of the population, they have a mean weight of 171kg with a standard deviation of 47kg. Their weights are assumed to be normally distributed. An elk is observed in the park, and found to have a weight of 95kg. Test, using the 5% significance level, whether this elk is underweight.
- An ice cream van driver starts a round on a new estate. The mean daily income she receives on this estate during her first month is £46.70 with a standard deviation of £14.30. At the beginning of a second month, she records the daily income from this estate of a random sample of 7 days. The total income for those 7 days is £450. Test, using the 5% significance level, whether the mean daily income from this estate exceeded £46.70. Assume that the sample of daily incomes is from a normal distribution with standard deviation 14.3.

6.4 Two-tailed tests

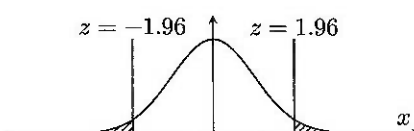
There may be occasions when we are looking for a value to be *close to the mean*, and not too far away in either direction.

Definition 6.6 - One-tailed and two-tailed tests

We use the term **one-tailed test** to refer to a hypothesis test involving restriction *in one direction only (upper tail or lower tail)*. If we need the value to be *between two limits* - not too far away from the mean in either direction - we use a **two-tailed test**, which studies the *middle of the distribution*.

For two-tailed tests, we need to calculate the critical values slightly differently - since we still want the result to have a *significance of 5%* we need to take into account that *half of this will be at either end* of the distribution, so we instead draw the lines at *97.5%* and *2.5%* (as we did for the 95% confidence interval).

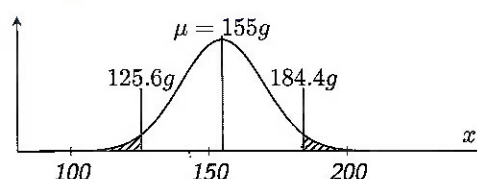
For a normal distribution, instead of 1.64, the critical value for the z-score will now be



Example 6.5.

Back at the widget factory from Example 6.3, the following week, a delegation from the We Make A Range Of Sizes of Widgets But Often Drop Things Factory has been to visit. A spare widget is discovered on the floor with a weight of *180g*. Did it fall off from the production line, or was it dropped by a visitor?

H_0 : widget was from production line
 H_1 : widget was dropped by a visitor.



For a significance level of 5%, we can use the *two-tailed z-score of 1.96*, and calculate

$$1.96 = \frac{x - \mu}{\sigma} = \frac{x - 155}{15}$$

$$1.96 \times 15 = x - 155 \Rightarrow (1.96 \times 15) + 155 = x \Rightarrow x = 184.4$$

This is the maximum weight for a result not to be considered significant. In the other direction, we can calculate:

$$155 - (184.4 - 155) = 125.6$$

So if the widget weighs between *125.6g and 184.4g* we can reject the alternative hypothesis and conclude

it was probably *from our production line*. Since our widget was 180g, we can *accept the null hypothesis*.

Exercise 6.3.

- (a) Research has shown that the mean recovery time for a cold is 18 days, with a standard deviation of 1.5 days, and follows a normal distribution. What are the upper and lower limits for the time taken to recover from a cold, to a significance level of 5%?
- (b) At a bakery, a machine is used to measure out quantities of dough for making loaves. For each batch, the mean amount of dough per load is set using a control on the machine. The weights of the resulting loaves are normally distributed with standard deviation 24 grams.
- The control is set to produce a batch of loaves with a mean weight of 1000 grams. The weights, in grams, of a random sample of nine loaves from this batch had a mean of 970.1 grams. Using the 5% significance level, examine whether the mean weight of this batch of loaves is different from 1000 grams.

6.5 Structure of a Hypothesis Test

The outline below may be helpful when approaching a hypothesis testing situation.

- Define your variables
- Formulate the hypotheses (H_0 and H_1)
- Declare the test including:
 - which tail/how many tails
 - the type of test
 - the significance level and
 - what you need to carry out the test
- Calculate the test statistic
- Determine the critical region OR Determine the p -value
- Compare the test statistic with the critical region OR Compare the p -value with the significance level
- Determine whether the result is significant
- Decide whether to reject H_0 or not
- Determine the conclusion.

Exercise 6.4. For each of the situations described below, follow the list above and suggest what you would write for the first three bullet points.

- (a) A flock of goldfinches eat seeds from a birdfeeder, and the amount of seed each bird consumes is normally distributed. A bird visits the feeder while nobody is watching, and you wish to determine if it was a goldfinch, or one of the larger birds sometimes seen in the area.
- (b) The heights of 10-year-old boys are normally distributed. A researcher wants to determine if a new experimental drug some boys were given as babies has affected their height.

Later we will see more examples of types of test which can be applied to different situations.